UNIVERSITY OF OTAGO EXAMINATIONS 2024

SCHOOL OF COMPUTING

COSC343

Artificial Intelligence

Semester Two

(TIME ALLOWED: 3 HOURS)

This examination paper comprises 6 pages

Candidates should answer questions as follows:

Candidates should answer **All** questions

Marks are shown thus: (2)

The total number of marks for this exam is 60

The following material is provided:

Nil

Use of calculators:

No calculators are permitted

Candidates are permitted copies of:

No additional material

Other Instructions:

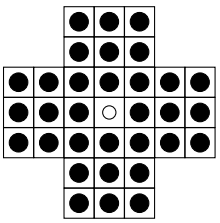
Nil

TURN OVER

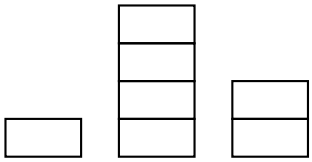
1. **State-space search**

Peg solitaire is a single-player game with the objective of removing all-but-one peg in the least number of moves from a grid of holes where one move is a jump of any chosen peg over an adjacent peg (removed in the process) into an empty hole.

Nim is a game of strategy where two players take turns removing (or "nimming") objects from distinct heaps or piles. On each turn, a player must remove at least one object, and may remove any number of objects provided they all come from the same heap or pile. The goal of the game is to avoid taking the last object.



Peg solitaire



Nim

(a) Given the general description of the two games above (i.e. you don't need to understand the details of the game mechanics) decide what type of state space-search technique would be suitable for an agent playing each game. In each case provide a general description of the space-search technique in question and how, in general terms, it informs the agent on what actions to take. (8)

(b) What is the role of the *transition model* in *look-ahead* search? (2)

(c) Propose a simple representation of the state of the Nim stacks shown in the image above. Explain your representation and how it can capture different states of the game. (2)

2. **Uncertainty**

The following information table about restaurants and Stuart Russel has four columns: how full a restaurant is; the type of restaurant; how hungry Stuart is; whether Stuart is willing to wait for a table given the other variables.

Patrons (Pa)	Type (T)	Hungry (H)	Will wait for table (W)
some	french	T	F
full	thai	T	F
some	burger	F	T
full	thai	T	T
some	french	F	F
some	italian	T	T
none	burger	F	F
some	thai	T	T
full	burger	F	F
full	italian	T	F
none	thai	F	F
full	burger	T	T

(a) According to the table above, what are the respective probabilities of waiting, $P(W = \mathrm{T})$, and not waiting, $P(W = \mathrm{F})$, for a table (fractions are fine)? (2)

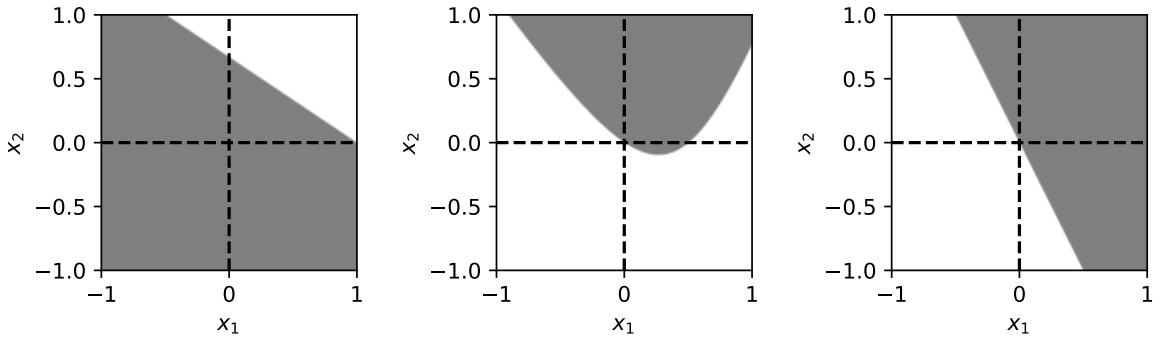
(b) What are the respective probabilities of a restaurant having *some* patrons given Stewarts's decision was to wait, $P(Pa = \mathrm{some}|W = \mathrm{T})$ and not wait, $P(Pa = \mathrm{some}|W = \mathrm{F})$, for a table? (2)

(c) What are the respective probabilities of Stuart being hungry given his decision to wait, $P(H = T|W = \mathrm{T})$ and not wait, $P(H = \mathrm{T}|W = \mathrm{F})$, for a table? (2)

(d) Suppose that we don't know the type of restaurant Stuart walked into, but we know it has *some* patrons and Stuart is hungry. Would a Naïve Bayes classifier predict it more likely that he will wait for a table or not? Show your calculations. Bayes formula, if you need it, is $P(A|B) = \dfrac{P(B|A)P(A)}{P(B)}$. (6)

3. **Classification and regression**

The three diagrams below depict different classifications over two-dimensional input space; the dark and white patches represent regions of the input space that are classified as belonging to class 1 and 0 respectively.



(i) (ii) (iii)

(a) Is a single-layer perceptron with a hard-limiting activation function capable of producing the classifications above? Answer yes or no for each case and justify your answer. (3)

(b) Is a single-layer perceptron with a hard-limiting activation function and a fixed bias value of zero capable of producing the classifications above? Answer yes or no for each case and justify your answer. (3)

(c) What, if any, architectural change would you need to introduce to the single-layer perceptron model to give it the capability of producing any of the three classifications above? What other, if any, changes would be needed to enable the learning of these classifications from data? (3)

(d) What is the difference between *regression* and *logistic regression* in terms of the hypothesis' architecture, training and applications? (3)

4. **Neural networks**

Consider an artificial neural network model with the output described by the following equation:

$$\hat{y} = \sigma_l\Big(\sigma_r(x \cdot W_1 + b_1) \cdot W_2 + b_2\Big),$$

where:

x is a $1 \times D$ vector of inputs,
W_1 is a $D \times U$ matrix,
W_2 is a $U \times K$ matrix,
b_1 is a U-dimensional vector,
b_2 is a K-dimensional vector,
$\hat{y}$ is a $1 \times K$ vector,

$$\sigma_l(v) = v,$$

$$\sigma_r(v) = \begin{cases} v & v > 0 \\ 0 & v \leq 0, \end{cases}$$

and $\cdot$ signifies the dot product.

(a) What type of neural network is this, how many layers does it have and what are the common names of the activation functions it uses? (2)

(b) Suppose you are to configure this network for processing 11 different meteorological measurements from a weather station for the purpose of predicting the temperature in the next hour. Choose appropriate values of D, U, and K. Explain in each case whether your choice is dictated by the problem/data or whether it's an arbitrary choice. (3)

(c) Compare and contrast the multilayer feed forward network architecture against the recurrent neural network architecture. Explain the similarities and differences in training and functionality. (3)

(d) Assume that meteorological measurements taken every hour over a 12-hour period are a good predictor of the temperature in the next hour. Is a feed-forward or recurrent architecture more suitable for this prediction? Justify your answer. (3)

(e) What does the term *universal function approximator* mean? (1)

5. **Unsupervised and reinforcement learning**

(a) Compare and contrast *supervised*, *unsupervised* and *reinforcement learning*. Provide an example application for each. (6)

(b) Explain the nature and objectives of *dimensionality reduction* and *clustering*? (2)

(c) The expression for utility of a reinforcement learning agent being in state s at time t is given by the following relation:

$$U(s_t) = r_t + \gamma U(s_{t+1})$$

What do the symbols r_t and γ represent and what role do they play in reinforcement learning? (3)

(d) How do Q-values in a Q-table relate to utility? (1)

END